Worksheet

Computer Network - Week 3 - DNS

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Activity 1 : nslookup

Screenshots

```
C:\Windows\system32>nslookup -type=NS iitb.ac.in
Server:  UnKnown
Address:  10.6.47.254

Non-authoritative answer:
iitb.ac.in      nameserver = dns3.iitb.ac.in
iitb.ac.in      nameserver = dns2.iitb.ac.in
iitb.ac.in      nameserver = dns1.iitb.ac.in

iitb.ac.in      nameserver = dns2.iitb.ac.in
iitb.ac.in      nameserver = dns1.iitb.ac.in
iitb.ac.in      nameserver = dns3.iitb.ac.in
dns3.iitb.ac.in internet address = 103.21.127.129
dns2.iitb.ac.in internet address = 103.21.126.129
dns1.iitb.ac.in internet address = 103.21.125.129
dns2.iitb.ac.in internet address = 103.21.126.129
dns1.iitb.ac.in internet address = 103.21.125.129
dns3.iitb.ac.in internet address = 103.21.127.129
```

Answers

1. What is the IP address of www.iitb.ac.in?

   10.6.47.254

2. What is the IP address of the DNS server that provided the answer to your nslookup command in

question 1 above?

103.21.125.129

3. Was the response from your nslookup command in question 1 authoritative or non-authoritative?

It is non-authorative

4. Use nslookup to determine the authoritative name server for the domain www.iitb.ac.in. What is its name? (If there is more than one authoritative server, what is the name of the first authoritative server returned by nslookup?) If you need to find the IP address of that authoritative name server, how would you do it?

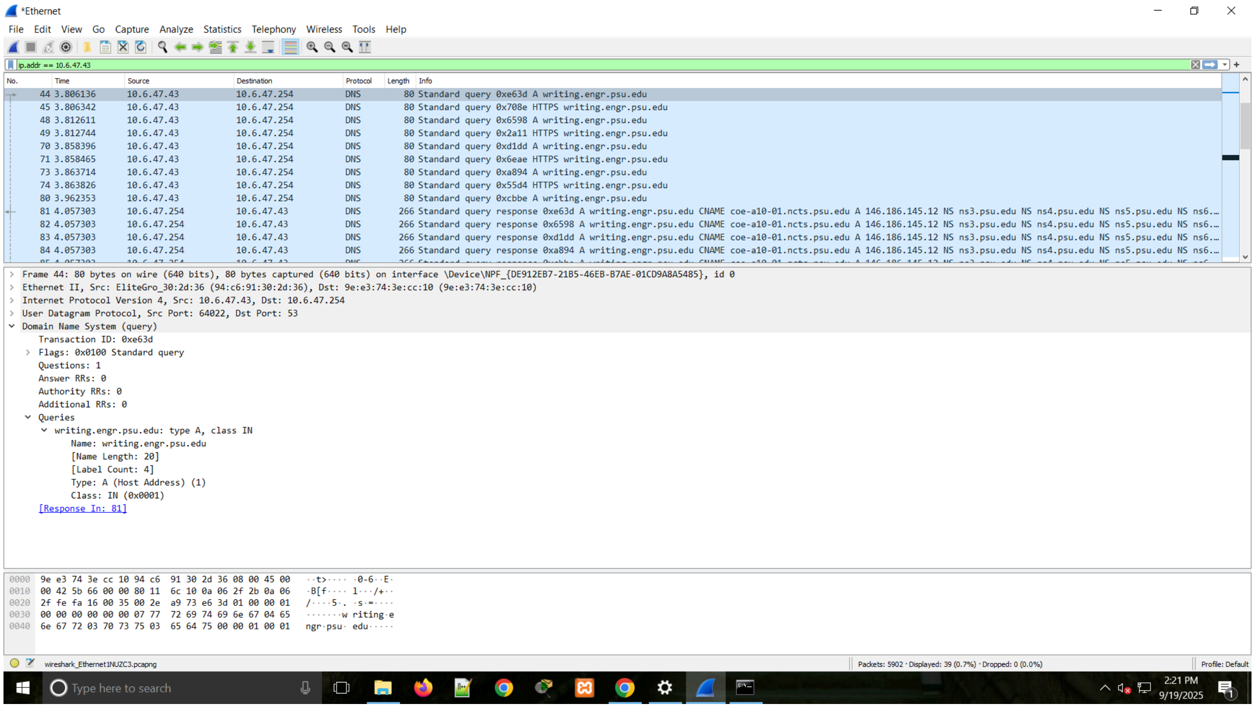
dns1.iitb.ac.in, in my command line, it is in the bottom line of the output, there are the name server with its IP address

Activity 2 : Flushing DNS Cache

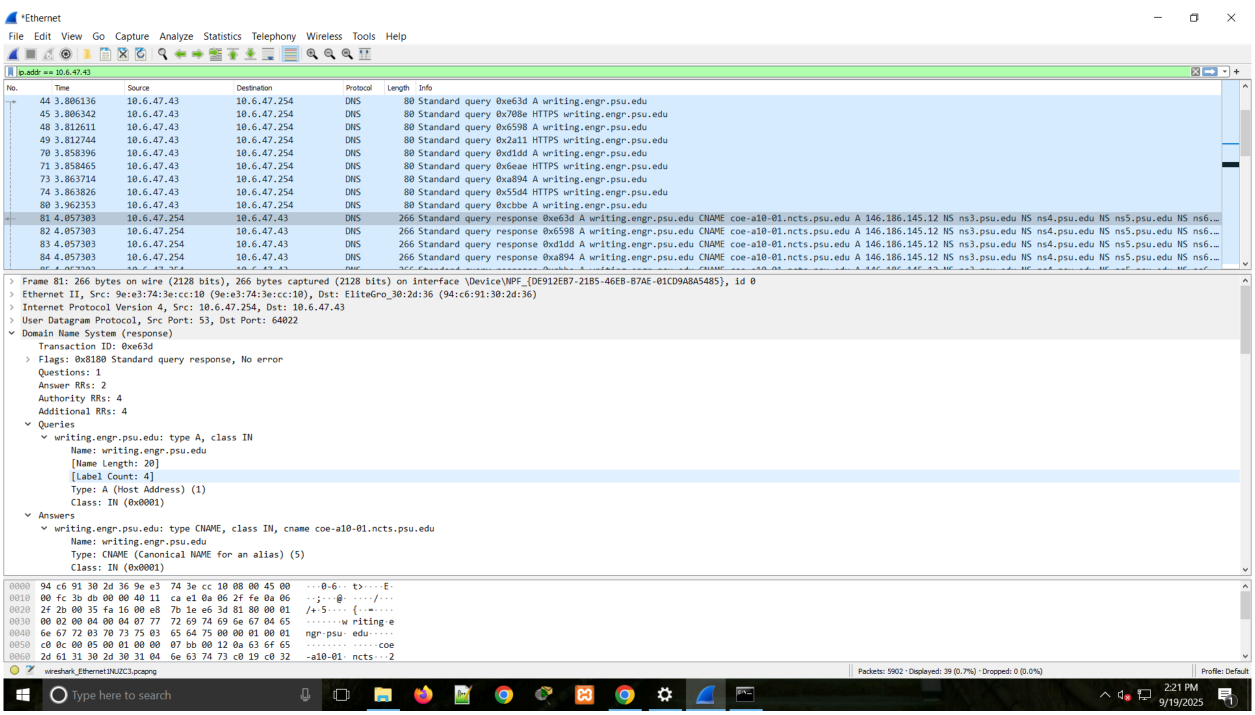
Screenshots

```
C:\Windows\system32>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::91e7:9651:c820:5862%4
   IPv4 Address. . . . . . . . . . . : 10.6.47.43
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.6.47.254

Tunnel adapter isatap.{DE912EB7-21B5-46EB-B7AE-01CD9A8A5485}:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Tunnel adapter Teredo Tunneling Pseudo-Interface:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2001:0:2851:782c:1068:38ef:f5f9:d0d4
   Link-local IPv6 Address . . . . . : fe80::1068:38ef:f5f9:d0d4%2
   Default Gateway . . . . . . . . . : ::
```

Ipconfig to check my IP address



Request message



Response message

Answers

1. To which IP address was the DNS request sent? Is this the IP address of your default local DNS server?

Yes, my ip address = 10.6.47.43

2. Check the DNS request message. What is the “Type” of this DNS request? Does the request message contain an “Answer”?

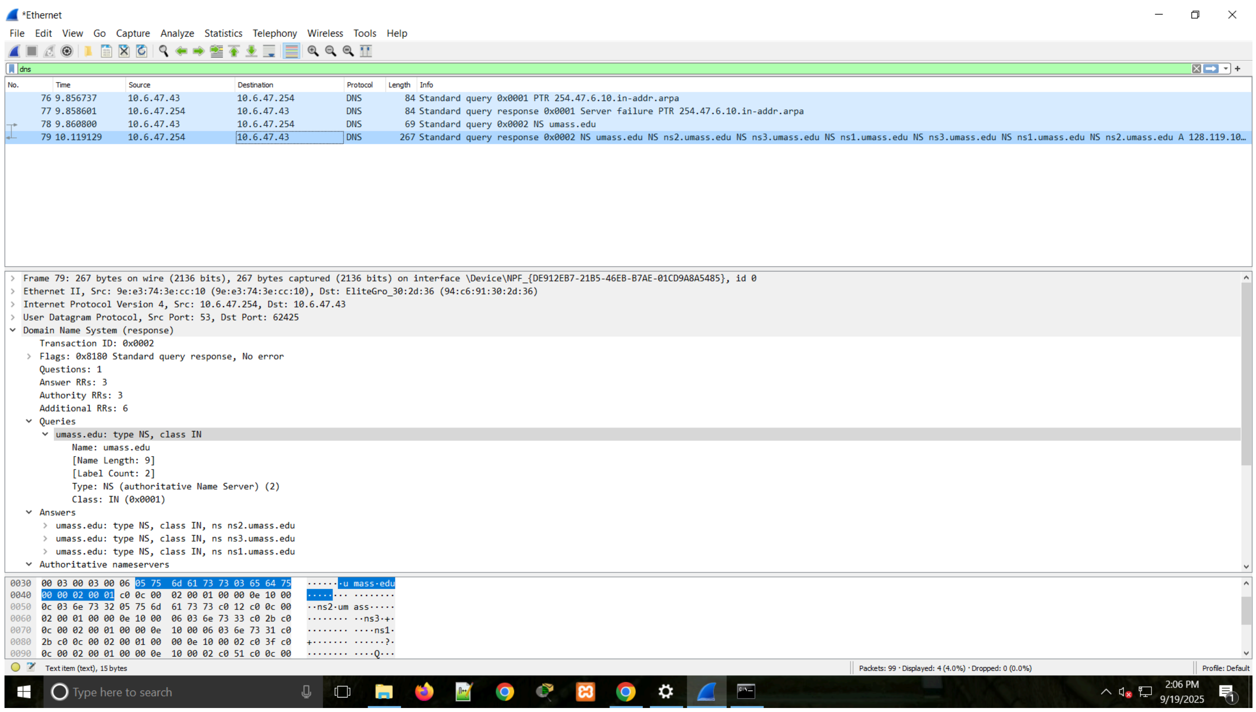
Type = A (Host Address), no it doesn’t contains the “answer”

3. Check the DNS response message for that request. How many “Questions” are included in the DNS response? How many “Answers”?

Questions : 1 Answer RRs : 2

Activity 3 : Capture & DNS Message Analysis with Wireshark

Screenshots



Answers

1. To which IP address was the DNS request sent? Is this the IP address of your default local DNS server?

Yes, my ip address = 10.6.47.43

2. Check the DNS request message. How many questions does the request contain? Does the request message include an “Answer”?

It has one “question”, yes it contain the “answer”

3. Check the DNS response message. How many answers does the response contain? What information is included in the answers? How many additional resource records are returned? What additional information is included in these additional resource records?

It has three answer; it contains the authoritative name server with its detail information like the name, type, class ; it returned six additional resource record; same like in “answer” but with its IP address